

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
KIRUMAMPAKKAM, PUDUCHERRY-607 403.

B.Tech., Ist Year Sem-I - PHYSICS – Model Exam-February 2023

Section: A,B&C
Date:11.02.2023

Max. Marks: 75
Time : 3 hrs

SECTION A (10x2=20marks)

Answer ALL the questions

All Questions carry equal marks

1. Define Reverberation time.
2. Define Doppler Effect.
3. In the Michelson interferometer arrangement, if one of the mirror is moved by 0.04mm, 125 fringes cross the field of view. Calculate the Wavelength of light used.
4. Define QWP and give its applications.
5. What are the characteristics features of a laser beam?
6. Explain the term acceptance angle of a fiber.
7. What are matter waves?
8. State Heisenberg uncertainty principle.
9. Define atomic mass unit.
10. Give the four essential components of a nuclear reactor.

SECTION B(5X11=55marks)

Answer FIVE questions in “Either or” Pattern

All questions carry equal marks

UNIT I

11. (a) What are X-cut and Y-cut crystal? (3)
(b) What is Inverse Piezo electric effect? Explain how the ultrasonic waves are produced using Piezo electric oscillator? (8)

OR

12. Write an essay on the factors affecting acoustics of buildings and give their remedies time (11)

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UNIT II

13. Explain the construction, working and types of fringes in Michelson interferometer and hence explain how it is used to find the wavelength of monochromatic source (11)

OR

14. a. Write a note on anti reflection coating. (4)
b. Explain briefly the resolving power of prism. (7)

UNIT III

15. Derive Einstein's coefficients A and B and hence explain the existence of stimulated emission. (11)

OR

16. Describe the, construction and working of a Nd-YAG laser with a suitable energy level diagram. (11)

UNIT IV

17. Derive time independent Schrödinger wave equation and hence deduce the time dependent Schrödinger wave equation. (11)

OR

18. Explain the application of Schrödinger wave equation using one dimensional potential well. (11)

UNIT V

19. With neat diagram explain the principle, construction, working merits and demerits of BWR. (11)

OR

20. With a neat sketch describe the construction and working of PWR (11)

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All the Best**